

Ex.No. 5

TEXT ANALYTICS USING tm AND SnowballC

```
library(tm)
```

```
library(SnowballC)
```

```
# Input text
```

```
text <- paste(
```

```
  "Students are learning text analytics using R programming language.",
```

```
  "R programming language is very tough for I MSc ADs B.",
```

```
  "Kindly change the syllabus."
```

```
)
```

```
# Step 1: Create corpus
```

```
corpus <- VCorpus(VectorSource(text))
```

```
# Step 2: Convert to lowercase
```

```
corpus <- tm_map(corpus, content_transformer(tolower))
```

```
# Step 3: Remove punctuation
```

```
corpus <- tm_map(corpus, removePunctuation)
```

```
# Step 4: Remove stopwords
```

```
corpus <- tm_map(corpus, removeWords, stopwords("english"))
```

Step 5: Remove extra spaces

```
corpus <- tm_map(corpus, stripWhitespace)
```

Step 6: Stemming

```
corpus <- tm_map(corpus, stemDocument)
```

Step 7: Create Term–Document Matrix

```
tdm <- TermDocumentMatrix(corpus)
```

Step 8: Display processed text

```
cat("Processed text:\n")
```

```
print(as.character(corpus[[1]]))
```

```
inspect(corpus)
```

Step 9: Display Term–Document Matrix

```
cat("\nTerm–Document Matrix\n")
```

```
inspect(tdm)
```